

CO5 – AT3 DAA MOCK INTERVIEW

Student Performance Report

Student Name	GAYATHERI S
Student ID	192521306
Course	Design and Analysis of Algorithms (DAA)
Interview Format	Technical Mock Interview
Interview Duration	Approximately 15 minutes
Number of Questions	10
Final Result	100 / 100 – Level 4

1. Introduction

This report presents the performance of GAYATHERI S in the CO5 – AT3 Design and Analysis of Algorithms (DAA) Mock Interview. The interview assessed technical knowledge, problem-solving ability, analytical thinking, communication, confidence, and response to questions. Ten questions were asked progressively across the specified DAA topics.

2. Topics Covered

- Backtracking and its comparison with brute-force search
- State-space trees and their nodes and branches
- Pruning and its role in improving efficiency
- N-Queens problem and queen attack conditions
- Subset Sum using include/exclude decisions
- P, NP, NP-Hard and NP-Complete complexity classes
- Polynomial-time reduction
- Exact versus heuristic algorithms
- Greedy algorithms and the greedy-choice principle
- Hamiltonian Cycle using backtracking

3. Question-wise Assessment

Q	Area	Assessment
1	Backtracking	Correctly explained incremental construction and abandoning invalid paths.
2	State-space Tree	Clearly explained states, root, nodes, and decision branches.
3	Pruning	Explained pruning as cutting invalid branches and reducing exponential search.
4	N-Queens	Correctly identified row, column, and diagonal conflicts.
5	Subset Sum	Clearly explained include/exclude branching, target checking, and pruning.
6	P/NP Classes	Strong explanation of P, NP, NP-Hard, and NP-Complete relationships.
7	Polynomial Reduction	Correctly described reduction direction and its use in NP-Hardness proofs.
8	Exact vs Heuristic	Clearly explained optimality versus speed with a routing example.
9	Greedy Algorithm	Correctly described local optimal choices and used coin change as an example.

Q	Area	Assessment
10	Hamiltonian Cycle	Strong explanation of adjacency, uniqueness, cycle closure, recursion, and backtracking.

4. Technical Knowledge

The student demonstrated strong command of the major concepts covered in the interview. The answers showed understanding of algorithmic mechanisms, constraints, complexity, pruning, recursive decision-making, and practical applications. The student was particularly strong in explaining how backtracking explores alternatives and how invalid branches are discarded.

5. Problem-Solving and Analytical Thinking

The student consistently explained not only what an algorithm does but also why a particular step is performed. Examples included the include/exclude structure for Subset Sum, diagonal checks in N-Queens, and validation conditions in Hamiltonian Cycle. These responses demonstrate strong analytical reasoning and understanding of constraints.

6. Communication and Confidence

The responses were organized, technically detailed, and confident. The student used appropriate algorithmic terminology and examples to support explanations. The main communication improvement is to make answers more concise because the interview format allows approximately 1–1.5 minutes per question.

7. Key Strengths

- Strong conceptual understanding of DAA topics.
- Accurate explanation of backtracking and pruning.
- Good understanding of NP-related complexity classes.
- Ability to connect theory with real-world examples.
- Clear reasoning for constraint-based problems.
- Strong explanation of recursive and decision-tree approaches.
- Consistent and confident responses throughout the interview.

8. Areas for Improvement

- Reduce the length of answers while preserving the key technical points.
- Prioritize the definition, core principle, and one example when time is limited.
- Avoid unrelated comments or references during technical responses.
- Practice delivering each answer naturally within approximately 60–90 seconds.
- Use concise interview-style explanations instead of giving every possible detail.

9. Evaluation Summary

Criterion	Weight	Level	Marks
Technical Knowledge	25	Level 4	25/25
Problem Solving	25	Level 4	25/25
Analytical Thinking	15	Level 4	15/15
Communication	15	Level 4	15/15
Confidence	10	Level 4	10/10

Response to Questions	10	Level 4	10/10
TOTAL	100	Level 4	100/100

10. Overall Conclusion

GAYATHERI S completed the DAA mock interview with an excellent overall performance. The student demonstrated Level 4 performance in all evaluation criteria and achieved a total of 100 out of 100. The technical responses were accurate, analytical, and supported by relevant examples. For further improvement, the student should focus mainly on concise delivery and time management during oral interviews.

11. Final Recommendation

Recommended preparation: Continue practicing DAA concepts using short 1-minute explanations, followed by a small example or complexity statement where appropriate. This will help maintain the current technical strength while making responses more effective in timed viva and interview settings.

Report prepared based on the CO5 – AT3 DAA Mock Interview session.